

Lab 2 ER Diagrams. Database Design and modelling tool

Exercise 1. Use draw.io to recreate the ER diagrams for the student-module-professor exercise discussed in class earlier. Pay special attention to the cardinality of relationships. You will create first a rough ER diagram with all entities, relationships and basic attributes and a second one including a more detailed list of attributes and all required entities. **See [video](#) for walk through.**

Exercise 2. Prepare an ER diagram for a real estate company that lists property for sale. The following describes the organisation:

- The company has a number of sales offices in several counties.
 - ➔ Attributes of sales office include Office_Number (identifier/key) and Location.
- Each sales office is assigned one or more employees.
 - ➔ Attributes of employee include Employee_ID (identifier/key) and Employee_Name.
 - ➔ An employee must be assigned to only one sales office.
- The firm lists property for sale.
 - ➔ Attributes of property include Property_ID (identifier) and Location.
 - ➔ Components of Location include Address, City, County, and Postal_Code.
- Each unit of property must be listed with one (and only one) of the sales offices.
 - ➔ A sales office may have any number of properties listed.
- Each unit of property has one or more owners.
 - ➔ Attributes of owners are Owner_ID (identifier) and Owner_Name.
 - ➔ An owner may own one or more units of property.

- ➔ An attribute of the association between property and owner is Percent_Owned.

Review

- Look at your diagram from the point of view of a system owner or user. Is everything clear? Discuss your diagram with students in your room.
- Check through the Cardinality pairs.
- Also, look over the list of attributes associated with each entity to see if anything has been omitted.

Exercise 3. Use draw.io to create the ER diagram from exercise 2.